

Toward Robust Machine Learning Models in Voice and Speech Science: A Hands-On Demonstration of Selection and Evaluation Biases

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1- Key Terms and Processes in Machine Learning

A. Key concepts

- **Artificial intelligence (AI):** Broad field focused on machines performing tasks that typically require human intelligence.
- **Machine learning (ML):** A subset of AI where systems learn patterns from data to make predictions or decisions.
- **Deep learning:** A specialized ML approach using neural networks with many layers, often for complex tasks like image or speech processing or recognition.
- **Generative AI:** Models that can create new content (e.g., text, images, audio) based on learned patterns. Current generative AI models are based on deep learning models.

B. Types of machine learning

- **Supervised learning:** The model is trained on data that includes both input features and known output labels. The goal is to learn a mapping from inputs to outputs so the model can make accurate predictions on new, unseen data.
- **Unsupervised learning:** The model is trained on data without labeled outcomes. The goal is to discover hidden patterns, groupings, or structures within the data.
- **Semi-supervised learning:** A machine learning approach that uses a small amount of labeled data along with a large amount of unlabeled data to train a model. It helps improve learning when labeling is expensive or time-consuming, by leveraging patterns in the unlabeled data to support the learning process.

C. Core terminology

- **Feature:** The inputs and attributes that help the model learn patterns and make predictions.
- **Feature extraction:** The process of transforming raw data into informative features.
- **Label:** The known outcome or category (e.g., patient vs. control) that a machine learning model is trained to predict.
- **Model:** The mathematical mapping from input features to output labels. It learns patterns from data during training and uses those patterns to make predictions or decisions on new data.
- **Dimensionality:** the number of input features used by a machine learning model. Higher dimensionality means the model is working with more variables, which can capture more detail but also increase the risk of overfitting if the sample size is small.
- **Sample size:** The number of data points or observations used to train and evaluate a machine learning model. Larger sample sizes generally lead to more reliable models, especially when working with high-dimensional data or complex tasks.
- **Sample size estimation:** The process of determining how much data is needed to train a model reliably and evaluate its performance with confidence. It depends on the discriminative power of the features, dimensionality, the number of extracted features, and the type of cross-validation used, among other factors.

D. Model development and evaluation process

- **Training:** The process by which a machine learning model learns patterns from data by adjusting its internal parameters to map input features to output labels. During training, the model uses known examples to minimize errors and improve its ability to make accurate predictions on new data.
- **Model parameters:** Internal values (like weights in a multiple linear regression) that a machine learning model learns during training to map input features to output labels.
- **Hyperparameter optimization:** The process of finding the best settings (called hyperparameters) for a machine learning model to improve its performance. Unlike model parameters, hyperparameters are set before training and control how the model learns.
- **Model selection:** The process of choosing the best machine learning model for a specific task based on how well different models perform. It involves comparing models using evaluation metrics like accuracy or error rate.
- **Feature selection:** The process of choosing the most important (i.e., the most discriminative) features for a machine learning model to improve accuracy and reduce complexity. It helps the model focus on the data that matters most.

- **Overfitting:** when a machine learning model learns the training data too well, including its noise, which makes it perform poorly on new, unseen data. It typically happens when the model is too complex relative to the amount or quality of data.
- **Cross-validation (CV):** A technique for evaluating how well a machine learning model generalizes to unseen data. It works by splitting the dataset into multiple subsets, training the model on some folds, and testing it on the remaining ones.
- **Nested CV:** An advanced CV technique necessary to get an unbiased estimate of the performance and to prevent data leakage in the presence of hyperparameter optimization, model selection, and feature selection.
- **Data leakage:** When information from outside the training dataset is accidentally used during model training. This leads to overly optimistic performance estimates and can cause the model to overfit, making it unreliable on truly unseen data.

2- Identifying Bias Throughout the AI Life Cycle

A. Problem-Framing and Study Design

- **Historical and systemic bias:** Existing inequities may be reflected in clinical tools, diagnostic criteria, or health records used to label data. [1]
- **Measurement bias:** Differences in recording devices, environments, or signal quality can introduce artifacts unrelated to the target variable.
- **Labelling bias:** Occurs when “ground-truth” labels encode subjective or inconsistent criteria (e.g., clinician ratings influenced by context or prior experience).

B. Data Acquisition

- **Sampling bias:** Occurs when the data used to train or test a model does not align with the true characteristics of the broader population.
- **Missing-data bias:** Systematic absence of certain sub-groups (e.g., individuals with severe impairment or limited technology access) can distort model learning.

C. Model Development and Evaluation

- **Algorithmic bias:** Some ML algorithms (e.g., decision trees) are more sensitive to imbalanced data distributions.
- **Data leakage:** Occurs when test information unintentionally informs model training, inflating accuracy.
- **Overfitting:** The model captures noise rather than meaningful signal, yielding poor performance on new data.

D. Deployment and Monitoring

- **Contextual drift:** Model performance may degrade as population characteristics, devices, or usage settings evolve.
- **Feedback-loop bias:** Clinical or user feedback influenced by prior model predictions can reinforce existing inequities.

3- Robust Data Collection Practices

A. Design for Alignment

- Engage community members from start to develop relevant research and promote buy-in.
- Define inclusion/exclusion criteria that reflect the full clinical and demographic diversity of the target population.
- Use stratified sampling or oversampling techniques to ensure adequate representation of minority groups. [2]
- Balance data collection across recording environments, devices, and geographic regions to capture real-world variability.

B. Standardize Procedures

- Employ consistent recording protocols and robust metadata documentation (e.g., device type, noise level, posture, language).
- Implement calibration routines and quality-control checks to minimize measurement bias.
- Use interoperable formats and FAIR data principles to support reproducibility and secondary analysis.

C. Mitigate Labeling Bias

- Train annotators using clearly defined, standardized rating guidelines.
- Utilize multiple independent raters and adjudication processes to improve reliability.
- Track inter-rater agreement (e.g., Cohen’s κ) and document disagreements for model auditing. [1]

D. Plan for Transparency and Auditability

- Document sampling rationale, missing-data handling, and data-processing steps.
- Incorporate stakeholder and participant feedback loops to identify gaps in data equity.
- Establish governance procedures for re-evaluation when expanding to new populations or modalities [3]

E. Evaluation and Power Considerations

- Use nested cross-validation to obtain unbiased performance estimates. [4]
- Conduct sample-size estimation tailored to model complexity, feature dimensionality, and desired statistical power.
- Report model performance with confidence intervals and stratified subgroup analyses to assess fairness.

4- The Effect of Different Evaluation Methodologies

Figure 1 presents four different CVs for evaluating performance of a supervised ML model.

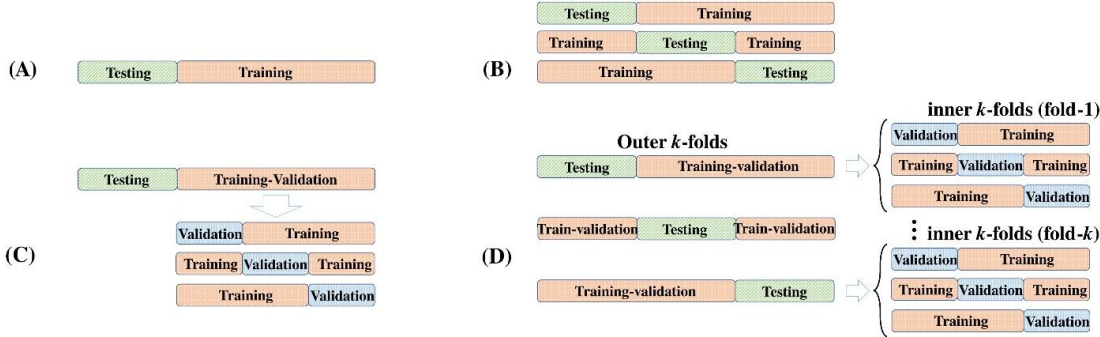


Figure 1. Visualization of data splitting for the four cross-validation methods studied: (A) single holdout, (B) k -fold ($k = 3$), (C) train-validation-test, (D) nested k -fold cross-validation ($k = 3$). [4]

An analysis was conducted to illustrate how the choice of CV can affect not only reported accuracy but also its stability and theoretical validity. The analysis used 48 features representing daily vocal function and behavior from individuals with phonotraumatic vocal hyperfunction and vocally healthy matched controls. [5] Figure 2(A) shows how average accuracy and its variability differ across CVs and feature dimensionality, while Figure 2(B) depicts the effect of sample size. The patterns reveal that simple approaches like single holdout often produce inflated and highly variable accuracy estimates, which become misleading as sample size decreases. In contrast, the more robust method of nested CV provides unbiased and consistent performance estimates, underscoring the importance of evaluation methodology, especially in studies with small sample sizes. [4]

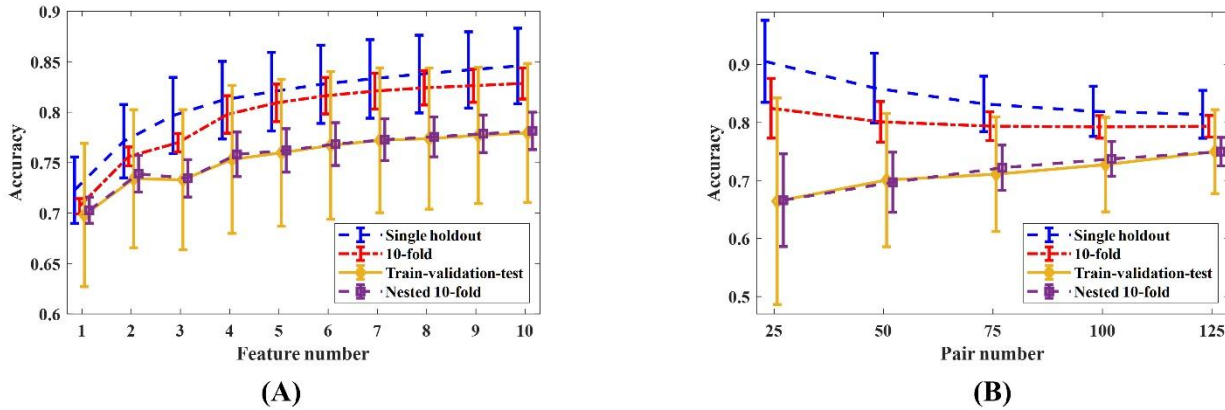


Figure 2. (A): The effect of different cross-validations on average and std of testing accuracy for different numbers of selected features, (B): the effect of sample size ($2 \times \text{pair number}$) on testing accuracy estimated based on different cross-validations. [1]

To further illustrate how CV can affect generalizability, we also measured the classifier's accuracy when discriminating between two sets of random, non-relevant features across different CV methods, in the presence of a feature selection component. The theoretical expected accuracy for such a balanced problem is 50%, making this analysis a good approach for detecting systematic bias. Figure 3 shows that popular methods like single holdout and standard k -fold exhibit significant bias, systematically overestimating accuracy and compromising the *validity* of the results. The figure also reveals large differences in variability among methods, indicating that the evaluation strategy also strongly impacts the *reliability* of ML outcomes. [1] Finally, we examined how different CVs affect the minimum sample size needed to train a statistically significant ML model. Single holdout required substantially more data to achieve conventional power ($\alpha = 0.05$, $1 - \beta = .8$). For instance, with two moderately discriminative features (effect size $d \approx 0.6$) in a 20-feature space, train-validation-test CV needed about 400 samples,

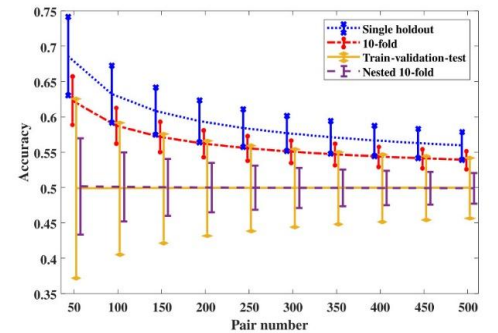


Figure 3. Mean and standard deviation of ML accuracy for non-relevant features (H_0) [1]

whereas nested 10-fold CV achieved the same power with only 200 samples. Therefore, a robust method can significantly reduce the required sample, making ML studies more practical when participant recruitment is costly and the sample size is small. [4]

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